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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2016 Half-Yearly Examination

FORM VI

MATHEMATICS EXTENSION 2

Thursday 25th February 2016

General Instructions

- Writing time — 1 hour 30 minutes
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 60 Marks

- All questions may be attempted.

Section I – 8 Marks

- Questions 1–8 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 52 Marks

- Questions 9–12 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Nine.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Reference sheet
- Candidature — 74 boys

Examiner

RCF

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

The expression $i^3 + i^6 + i^9 + i^{12} + i^{15}$ is equal to:

1

- (A) $-i$
- (B) i
- (C) 1
- (D) -1

QUESTION TWO

Suppose $z = r(\cos \theta + i \sin \theta)$ is a complex number. The complex number z^2 will be:

1

- (A) $r^2(\cos \theta^2 + i \sin \theta^2)$
- (B) $2r(\cos 2\theta + i \sin 2\theta)$
- (C) $r^2(\cos 2\theta + i \sin 2\theta)$
- (D) $2r(\cos \theta^2 + i \sin \theta^2)$

QUESTION THREE

What are the solutions to the quadratic equation $(z - 2 - i)(z + 3 + 2i) = 0$?

1

- (A) $z = 2 - i$ or $-3 + 2i$
- (B) $z = -2 - i$ or $3 + 2i$
- (C) $z = 2 + i$ or $-3 - 2i$
- (D) $z = -2 + i$ or $3 - 2i$

QUESTION FOUR

Which of the following is a primitive of $\frac{e^{\sqrt{x}}}{\sqrt{x}}$?

1

- (A) $2e^{\sqrt{x}}$
- (B) $(e^{\sqrt{x}})^2$
- (C) $\ln(e^{-\sqrt{x}})$
- (D) $\sqrt{x}e^{\sqrt{x}}$

Exam continues next page ...

QUESTION FIVE

Consider the function $f(x) = 2 \cos^{-1} \left(\frac{1-x}{2} \right)$. What is the gradient of the graph $y = f(x)$ at its y -intercept? **1**

(A) $-\frac{4}{\sqrt{3}}$

(B) $\frac{4}{\sqrt{3}}$

(C) $-\frac{2}{\sqrt{3}}$

(D) $\frac{2}{\sqrt{3}}$

QUESTION SIX

A primitive of $2 \cot x$ is:

1

(A) $\frac{1}{2} \ln(\cos^2 x)$

(B) $2 \ln(\cos x)$

(C) $\frac{-2}{\sin^2 x}$

(D) $2 \ln(\sin x)$

QUESTION SEVEN

The Kappa curve has the Cartesian equation

1

$$x^2 y^2 + y^4 = a^2 x^2$$

where a is a known constant. Which of the following is $\frac{dy}{dx}$?

(A) $\frac{-xy}{x^2 + 2y^2}$

(B) $\frac{x(a^2 - y^2)}{y(x^2 + 2y^2)}$

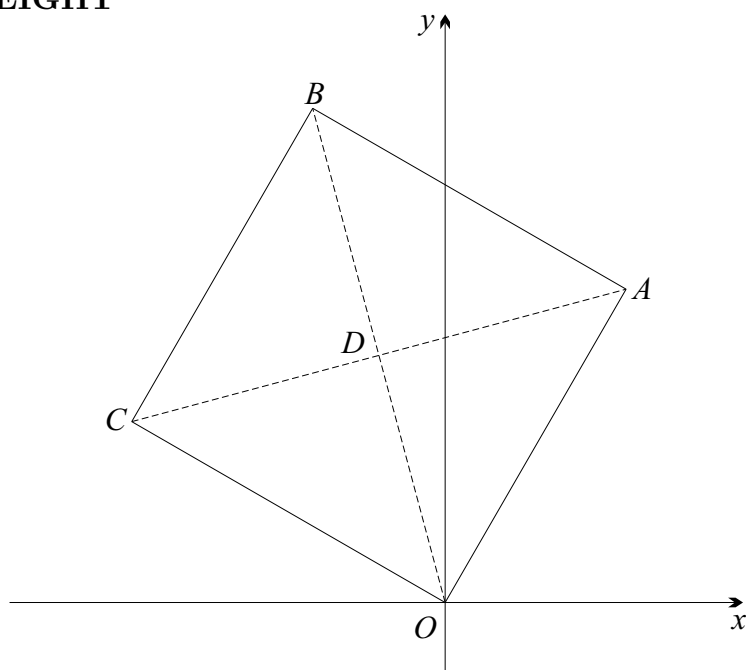
(C) $\frac{2(a^2 - y^2)}{xy(x^2 + 2y^2)}$

(D) $\frac{x(a^2 - y^2)}{y^3}$

Exam continues overleaf ...

QUESTION EIGHT

1



The diagram above shows a square $OABC$ in the complex plane. The vertex A represents the complex number $1 + i\sqrt{3}$. Which of the following complex numbers represents point D , the intersection of the diagonals?

- (A) $\left(\frac{1 - \sqrt{3}}{2}\right) + \left(\frac{\sqrt{3} - 1}{2}\right)i$
- (B) $\left(\frac{1 - \sqrt{3}}{2}\right) + \left(\frac{1 + \sqrt{3}}{2}\right)i$
- (C) $\left(\frac{1 + \sqrt{3}}{2}\right) + \left(\frac{\sqrt{3} - 1}{2}\right)i$
- (D) $\left(\frac{1 + \sqrt{3}}{2}\right) + \left(\frac{1 - \sqrt{3}}{2}\right)i$

_____ End of Section I _____

Exam continues next page ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION NINE (13 marks) Use a separate writing booklet.

Marks

- (a) Let $w = 1 + 3i$ and $z = 2 - 2i$.
- (i) Evaluate $i^3 z + z$. 1
 - (ii) Find $\overline{w - z}$. 1
 - (iii) Evaluate $|w + z|^2$. 1
 - (iv) Find $\operatorname{Re}(wz)$. 1
 - (v) Express $\frac{z}{w}$ in the form $a + ib$, where a and b are real numbers in simplest form. 2
- (b) (i) Express $u = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$ and $v = 1 - i\sqrt{3}$ in modulus-argument form. 1
- (ii) Hence, or otherwise, find $\arg(u^2 v)$. 1
- (c) Sketch the locus of those points z in the complex plane such that $|z - i| = |z + 1|$. 1
- (d) (i) Find the two square roots of $-7 + 24i$. 2
- (ii) Hence, or otherwise, solve the quadratic equation $z^2 + (1 + 2i)z + (1 - 5i) = 0$. 2

QUESTION TEN (13 marks) Use a separate writing booklet.

Marks

(a) (i) Find $\int \frac{x}{25 - 4x^2} dx$. 1

(ii) Find $\int \frac{x}{(25 - 4x^2)^3} dx$. 1

(iii) Find $\int \frac{1}{\sqrt{25 - 4x^2}} dx$. 2

(b) Use partial fractions to find $\int \frac{x + 1}{x^2 + 4x - 21} dx$. 3

(c) (i) Use the compound angle formulae to show that 1

$$\cos \alpha \cos \beta = \frac{1}{2} \cos(\alpha + \beta) + \frac{1}{2} \cos(\alpha - \beta).$$

(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \cos 6x \cos 5x dx$. 2

(d) Use integration by parts to evaluate $\int_0^1 4xe^{2x} dx$. 3

QUESTION ELEVEN (13 marks) Use a separate writing booklet.

Marks

(a) (i) Show that $\frac{d}{du} \log_e (u + \sqrt{a^2 + u^2}) = \frac{1}{\sqrt{a^2 + u^2}}$, where a is a constant. **1**

(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{\cos x}{\sqrt{4 + \sin^2 x}} dx$. **1**

(b) Use the substitution $t = \tan \frac{x}{2}$ to find $\int \frac{1}{1 + \cos x + \sin x} dx$. **3**

(c) A complex number satisfies $|z - 4| \leq 2$ and $\text{Im}(z) \leq 0$.

(i) Sketch the locus of z . **2**

(ii) Show that $-\frac{\pi}{6} \leq \arg z \leq 0$. **1**

(d) By rationalising the numerator of the integrand, evaluate $\int_{\frac{1}{2}}^1 \sqrt{\frac{x}{2-x}} dx$. **2**

(e) Use a suitable substitution to evaluate $\int_3^9 \frac{3}{(9+x)\sqrt{x}} dx$. **3**

QUESTION TWELVE (13 marks) Use a separate writing booklet.**Marks**

(a) Determine the locus specified by $\operatorname{Im}\left(z + \frac{1}{z}\right) = 0$.

3

(b) Let $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$.

(i) Prove that $I_n = n\left(\frac{\pi}{2}\right)^{n-1} - n(n-1)I_{n-2}$ for $n \geq 2$.

2

(ii) Hence evaluate $I_4 = \int_0^{\frac{\pi}{2}} x^4 \sin x \, dx$.

2

(c) Find $\int \sec^5 x \, dx$.

3

You may use the fact that $\int \sec^3 x \, dx = \frac{1}{2}(\sec x \tan x + \log(\sec x + \tan x)) + C$.

(d) In the Fibonacci sequence

3

$$1, 1, 2, 3, 5, 8, \dots$$

the terms of the sequence are defined recursively by the equation

$$F_{n+1} = F_n + F_{n-1}$$

where $F_1 = 1, F_2 = 1$ and $n \geq 2$.

Use Induction to prove that, for $n \geq 2$,

$$F_n^2 - F_{n+1}F_{n-1} = (-1)^{n+1}.$$

————— End of Section II —————

END OF EXAMINATION

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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

EXTENSION 2 H-Y 2016

$$\textcircled{1} i^3 + i^6 + i^9 + i^{12} + i^{15} = (-i) + (-1) + i + 1 + (-i) = (-i) \quad (\text{A}) \checkmark$$

$$\textcircled{2} z^2 = r^2 \cos 2\theta + i \sin 2\theta \quad (\text{C}) \checkmark$$

$$\textcircled{3} (z - (2+i))(z - (-3-2i)) = 0 \quad (\text{C}) \checkmark$$

$\therefore z = 2+i \text{ or } -3-2i$

$$\textcircled{4} \frac{d}{dx}(e^{\sqrt{x}}) = \frac{1}{2\sqrt{x}} e^{\sqrt{x}} \therefore F(x) = 2e^{\sqrt{x}} \quad (\text{A}) \checkmark$$

$$\textcircled{5} y = 2\cos^{-1}\left(\frac{1-x}{2}\right)$$

$$\frac{dy}{dx} = 2 \left(-\frac{1}{\sqrt{1-\left(\frac{1-x}{2}\right)^2}} \times \left(-\frac{1}{2}\right) \right)$$

$$= \frac{1}{\sqrt{1-\left(\frac{1-x}{2}\right)^2}}$$

$$\left(\frac{dy}{dx}\right)_{x=0} = \frac{1}{\sqrt{1-\frac{1}{4}}} = \frac{2}{\sqrt{3}} \quad (\text{D}) \checkmark$$

$$\textcircled{6} \int \frac{2\cos x}{\sin x} dx = 2\ln(\sin x) \quad (\text{D}) \checkmark$$

$$\textcircled{7} xy^2 + y^4 = x^2$$

$$\textcircled{2x} \quad 2xy^2 + x^2 2y \frac{dy}{dx} + 4y^3 \frac{dy}{dx} = 2x$$

$$= \frac{dy}{dx} (2xy^2 + 4y^3) = 2x - 2xy^2$$

$$\frac{dy}{dx} = \frac{2x - 2xy^2}{2xy^2 + 4y^3}$$

$$= \frac{x(1-y^2)}{y(x+2y^2)} \quad (\text{B}) \checkmark$$

$\textcircled{8}$ A represents $z = 1+i\sqrt{3}$
 C represents $iz = -\sqrt{3}+i$ (Rotation anticlockwise of 90°)
 B represents $z+iz = (1-\sqrt{3})+i(1+\sqrt{3})$
 D represents $\frac{1}{2}(z+iz) = \left(\frac{1-\sqrt{3}}{2}\right) + i\left(\frac{1+\sqrt{3}}{2}\right) \quad (\text{B}) \checkmark$

Question 9

a) $w = 1+3i \quad z = 2-2i$

$$\textcircled{i} i^3 z + z = (-i+1)z = (-i+1)(2-2i) = -2i+2i^2-2i+2 = -2i+2i^2-2i+2 = -4i \checkmark$$

OR $-i(2-2i) + (2-2i) = -2i+2i^2+2-2i = -2i-2+2-2i = -4i$

$$\textcircled{ii} \overline{w-z} = \overline{(1+3i)-(2-2i)} = \overline{-1+5i} = -1-5i \checkmark$$

$$\textcircled{iii} |w+z|^2 = |3+i|^2 = (3+i)(3-i) = 9+3i-3i-i^2 = 9+1 = 10 \checkmark$$

$$\textcircled{iv} \operatorname{Re}(wz) = \operatorname{Re}\{(1+3i)(2-2i)\} = \operatorname{Re}\{2+6i-2i+6\} = \operatorname{Re}\{8+4i\} = 8 \checkmark$$

$$\textcircled{v} \frac{z}{w} = \frac{2-2i}{1+3i} \times \frac{1-3i}{1-3i} = \frac{2-2i-6i-6}{1+9} = \frac{-4-8i}{10} \checkmark$$

$= \frac{-4-8i}{10} \quad \text{ie } a = \left(-\frac{2}{5}\right) \quad b = \left(-\frac{4}{5}\right)$
 $= \frac{-(2+4i)}{5} = \left(-\frac{2}{5}\right) + \left(-\frac{4}{5}\right)i$

Multi Choice Summary

$\textcircled{1} \text{A} \textcircled{2} \text{C} \textcircled{3} \text{C}$

$\textcircled{4} \text{A} \textcircled{5} \text{D} \textcircled{6} \text{D}$

$\textcircled{7} \text{B} \textcircled{8} \text{B}$

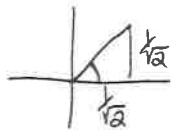
(1 mark each)

$$b) (i) u = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$$

$$\Rightarrow r=1$$

$$\theta = \frac{\pi}{4}$$

$$u = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4}$$



$$v = 1 - i\sqrt{3}$$

$$\Rightarrow r=2$$

$$\theta = (-\frac{\pi}{3})$$

$$v = 2 \cos(-\frac{\pi}{3}) + i \sin(-\frac{\pi}{3})$$

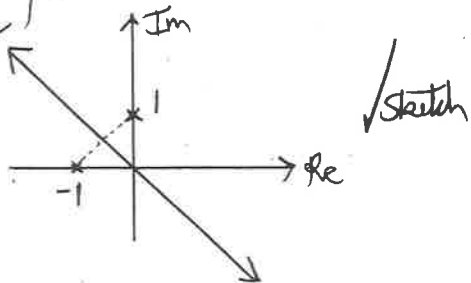


$$(ii) \arg(uv) = 2 \times \frac{\pi}{4} + (-\frac{\pi}{3})$$

$$= \frac{\pi}{2} - \frac{\pi}{3}$$

$$= \frac{\pi}{6}$$

c) Locus of points equidistant from $(-1,0)$ and $(0,1)$ is perpendicular bisector



$$d) (i) (a+ib)^2 = -7+24i$$

$$a^2 - b^2 + 2abi = -7 + 24i$$

$$a^2 - b^2 = -7$$

$$ab = 12$$

$$\therefore b=4, a=3 \text{ or } b=(-4), a=(-3)$$

square roots are $3+4i$ and $-(3+4i)$

$$(ii) z^2 + (1+2i)z + (1-5i) = 0$$

$$\Delta = (1+2i)^2 - 4 \times 1 \times (1-5i)$$

$$= 1 - 4 + 4i - 4 + 20i$$

$$= -7 + 24i$$

$$z = \frac{-(1+2i) \pm \sqrt{-7+24i}}{2}$$

$$= \frac{2+2i}{2} = 1+i \text{ or } \frac{-4-6i}{2} = -2-3i$$

$$(10) a) (i) \int \frac{x}{25-4x^2} dx = -\frac{1}{8} \ln |25-4x^2| + C$$

$$(ii) \int \frac{x}{(25-4x^2)^3} dx = \frac{1}{16} (25-4x^2)^{-2} + C$$

$$= \frac{1}{16(25-4x^2)^2} + C$$

$$(iii) \int \frac{1}{\sqrt{25-4x^2}} = \frac{1}{2} \int \frac{1}{\sqrt{\frac{25}{4}-x^2}} dx$$

$$= \frac{1}{2} \sin^{-1} \left(\frac{x}{5/2} \right) + C$$

$$= \frac{1}{2} \sin^{-1} \left(\frac{2x}{5} \right) + C$$

Do not penalize omission of constant

$\sqrt{1} \rightarrow \frac{2x}{5}$
 $\rightarrow$ Coeff correct

$$b) \int \frac{x+1}{x^2+4x-21} dx = \int \frac{x+1}{(x+7)(x-3)} dx$$

Let $\frac{x+1}{(x+7)(x-3)} = \frac{A}{x+7} + \frac{B}{x-3}$

$$= \int \frac{3/5}{x+7} + \frac{2/5}{x-3} dx$$

By Cover Up Rule

$$4 = 10B \Rightarrow B = \frac{2}{5}$$

$$-6 = -10A \Rightarrow A = \frac{3}{5}$$

$$= \frac{3}{5} \ln(x+7) + \frac{2}{5} \ln(x-3) + C$$

$$= \frac{1}{5} [3 \ln(x+7) + 2 \ln(x-3)] + C$$

$$c) (i) \cos(x+\beta) = \cos x \cos \beta - \sin x \sin \beta \quad (1)$$

$$\cos(x-\beta) = \cos x \cos \beta + \sin x \sin \beta \quad (2)$$

$$(1)+(2) \quad \cos(x+\beta) + \cos(x-\beta) = 2 \cos x \cos \beta$$

$$\frac{1}{2} [\cos(x+\beta) + \cos(x-\beta)] = \cos x \cos \beta$$

✓ "Show"

$$\begin{aligned}
 \text{ii) } \int_0^{\pi/2} \cos 6x \cos 5x \, dx &= \int_0^{\pi/2} \frac{1}{2} (\cos 11x + \cos x) \, dx \\
 &= \frac{1}{2} \left[\frac{1}{11} \sin 11x + \sin x \right]_0^{\pi/2} \checkmark \\
 &= \frac{1}{2} \left[\left(\frac{1}{11} \sin \frac{11\pi}{2} + \sin \frac{\pi}{2} \right) - 0 \right] \\
 &= \frac{1}{2} \left(1 - \frac{1}{11} \right) \\
 &= \frac{5}{11} \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{d) } \int_0^1 4xe^{2x} \, dx & \quad u=4x \quad \frac{dv}{dx}=e^{2x} \\
 &= \left[4x \left(\frac{1}{2} e^{2x} \right) \right]_0^1 - \int_0^1 4 \cdot \frac{1}{2} e^{2x} \, dx \quad \frac{du}{dx}=4 \quad v=\frac{1}{2} e^{2x} \\
 &= \left[2xe^{2x} - e^{2x} \right]_0^1 \checkmark \\
 &= \left[(2x-1)e^{2x} \right]_0^1 = e^2 - (-1e^0) = e^2 + 1 \checkmark
 \end{aligned}$$

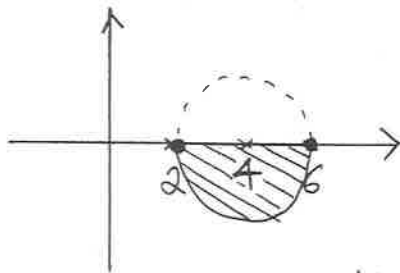
$$\begin{aligned}
 \text{⑪ a) (i) } \frac{d}{du} \ln(u + \sqrt{a^2 + u^2}) &= \frac{1}{u + \sqrt{a^2 + u^2}} \times \left(1 + \frac{1}{2}(a^2 + u^2)^{-\frac{1}{2}} \times 2u \right) \\
 &= \frac{1 + \frac{u}{\sqrt{a^2 + u^2}}}{u + \sqrt{a^2 + u^2}} \times \frac{\sqrt{a^2 + u^2}}{\sqrt{a^2 + u^2}} \quad \checkmark \text{ "Show"} \\
 &= \frac{(\sqrt{a^2 + u^2} + u)}{(u + \sqrt{a^2 + u^2}) \sqrt{a^2 + u^2}} \\
 &= \frac{1}{\sqrt{a^2 + u^2}}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } \int_0^{\pi/2} \frac{\cos x}{\sqrt{4 + \sin^2 x}} \, dx & \quad \text{in } a=2 \quad u=\sin x \quad \frac{du}{dx}=\cos x \\
 &= \left[\ln(\sin x + \sqrt{4 + \sin^2 x}) \right]_0^{\pi/2} \\
 &= \ln(1 + \sqrt{5}) - \ln(0 + 2) \\
 &= \ln\left(\frac{1 + \sqrt{5}}{2}\right) \checkmark
 \end{aligned}$$

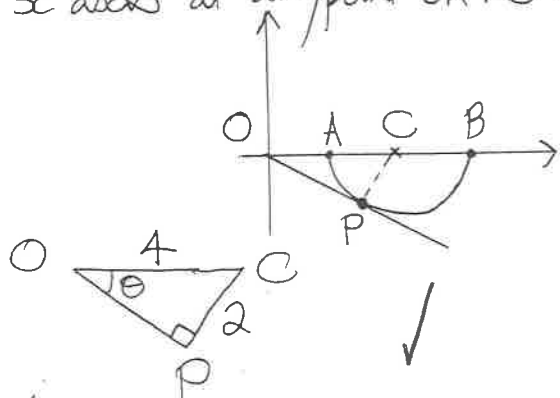
$$\begin{aligned}
 \text{b) } I &= \int \frac{1}{1 + \cos x + \sin x} \, dx \quad \text{Let } t = \tan\left(\frac{x}{2}\right) \\
 &= \int \frac{1}{1 + \left(\frac{1-t^2}{1+t^2}\right) + \left(\frac{2t}{1+t^2}\right)} \, dx \quad \checkmark \\
 & \quad \left(\times \frac{1+t^2}{1+t^2} \right) \\
 & \quad \frac{dt}{dx} = \frac{1}{2} \sec^2\left(\frac{x}{2}\right) \\
 & \quad = \frac{1}{2} (1 + \tan^2\left(\frac{x}{2}\right)) \\
 & \quad \therefore \frac{dt}{dx} = \frac{1}{2} (1 + t^2) \\
 & \quad dx = \frac{2dt}{1+t^2} \\
 & \quad \cos x = \frac{1-t^2}{1+t^2} \quad \sin x = \frac{2t}{1+t^2}
 \end{aligned}$$

$$\begin{aligned}
 &= \int \frac{2dt}{1+t^2+1-t^2+2t} \\
 &= \int \frac{2dt}{2+2t} \quad \checkmark \\
 &= \int \frac{dt}{1+t} \\
 &= \ln|1+t| + C \\
 &= \ln|1+\tan \frac{x}{2}| + C \quad \checkmark
 \end{aligned}$$

c) (i) $|z-4| \leq 2$
 Inside circle
 centred at $(4,0)$ $\checkmark$
 radius 2
 x-axis and $\checkmark$
 below



(ii) Minimum and maximum arguments occur for
 P, at tangent from origin in 4th quad, and
 on x-axis at any point on the interval AB.



$$\begin{aligned}
 \sin \theta &= \frac{2}{4} \\
 \sin \theta &= \frac{1}{2} \\
 \theta &= \frac{\pi}{6}
 \end{aligned}$$

$$\therefore -\frac{\pi}{6} \leq \arg z \leq 0$$

$$\begin{aligned}
 d) \int_{\frac{1}{2}}^1 \sqrt{\frac{x}{2-x}} dx &\quad \left(\frac{x\sqrt{x}}{x\sqrt{x}} \right) = \int_{\frac{1}{2}}^1 \frac{x}{\sqrt{2x-x^2}} dx \\
 &= \int_{\frac{1}{2}}^1 \frac{1-(1-x)}{\sqrt{2x-x^2}} dx \\
 &= \int_{\frac{1}{2}}^1 \frac{1}{\sqrt{1-(1-x)^2}} dx - \frac{1}{2} \int_{\frac{1}{2}}^1 \frac{2(1-x)}{(2x-x^2)^{\frac{3}{2}}} dx \quad \checkmark \\
 &= \left[-\sin^{-1}(1-x) - \sqrt{2x-x^2} \right]_{\frac{1}{2}}^1 \\
 &= \left(-\sin^{-1} 0 - \sqrt{1} \right) - \left(-\sin^{-1} \frac{1}{2} - \sqrt{\frac{3}{4}} \right) \\
 &= -1 + \frac{\pi}{6} + \frac{\sqrt{3}}{2} \quad \checkmark \\
 &= \frac{\pi + 3\sqrt{3} - 6}{6}
 \end{aligned}$$

$$\begin{aligned}
 e) I &= \int_3^9 \frac{3}{(9+x)\sqrt{x}} dx \quad \text{Let } u = \sqrt{x} \\
 &\quad du = \frac{1}{2\sqrt{x}} dx \\
 &= \int_{\sqrt{3}}^3 \frac{3 \times 2 du}{9+u^2} \quad \checkmark \quad \begin{array}{c|c|c} x & 3 & 9 \\ \hline u & \sqrt{3} & 3 \end{array} \\
 &= 6 \int_{\sqrt{3}}^3 \frac{du}{9+u^2} \\
 &= 6 \left[\frac{1}{3} \tan^{-1} \frac{u}{3} \right]_{\sqrt{3}}^3 \quad \checkmark \\
 &= 2 \tan^{-1} 1 - 2 \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) \\
 &= \frac{\pi}{2} - \frac{\pi}{3} \\
 &= \frac{\pi}{6} \quad \checkmark
 \end{aligned}$$

12) a) $\text{Im}(Z + \frac{1}{Z}) = 0$

Domain
 $Z \neq 0$

Let $Z = x + iy$.

$$Z + \frac{1}{Z} = x + iy + \frac{1}{x + iy}$$

$$= x + iy + \frac{(x - iy)}{x^2 + y^2}$$

$$= \frac{x^3 + xy^2 + ix^2y + iy^3 + x - iy}{x^2 + y^2}$$

$$= \frac{(x^3 + xy^2 + x) + i(x^2y + y^3 - y)}{x^2 + y^2}$$

A neat alternative which
simplifies algebra is:
 $\text{Im}(Z) + \text{Im}(\frac{1}{Z})$

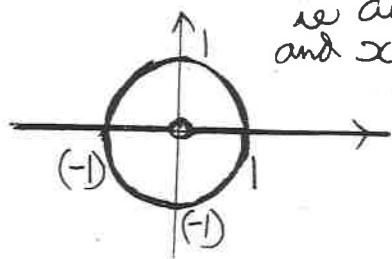
(NB $x^2 + y^2 \neq 0$
since $Z \neq 0$)

$$\therefore \text{Im}(Z + \frac{1}{Z}) = 0 \Rightarrow x^2y + y^3 - y = 0$$

$$y(x^2 + y^2 - 1) = 0$$

$$\therefore y = 0 \text{ OR } x^2 + y^2 = 1$$

ie circle radius 1 centre 0
and x axis excluding
origin.



Circle ✓
Line excluding
origin ✓

b) $I_n = \int_0^{\pi/2} x^n \sin x \, dx$ By Parts $u = x^n$ $\frac{du}{dx} = nx^{n-1}$ $\frac{dv}{dx} = \sin x$ $v = -\cos x$

$$= [-\cos x \times x^n]_0^{\pi/2} - \int_0^{\pi/2} -\cos x (nx^{n-1}) \, dx$$

$$= 0 + n \int_0^{\pi/2} x^{n-1} \cos x \, dx$$

By Parts $u = x^{n-1}$ $\frac{du}{dx} = (n-1)x^{n-2}$ $\frac{dv}{dx} = \cos x$ $v = \sin x$

$$= n \left\{ [x^{n-1} \sin x]_0^{\pi/2} - \int_0^{\pi/2} (n-1)x^{n-2} \sin x \, dx \right\}$$

$$= n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) I_{n-2}$$

as required

(ii) $I_0 = \int_0^{\pi/2} \sin x \, dx$

$$= [-\cos x]_0^{\pi/2}$$

$$= -\cos \frac{\pi}{2} - (-\cos 0)$$

$$= 1$$

$$I_2 = 2 \left(\frac{\pi}{2} \right) - 2 \times 1 \times I_0$$

$$= \pi - 2$$

$$I_4 = 4 \left(\frac{\pi}{2} \right)^3 - 4 \times 3 I_2$$

$$= \frac{\pi^3}{2} - 12\pi + 24$$

c) $\int \sec^5 x \, dx = \int \sec^3 x \times \sec^2 x \, dx$

Let $\frac{dv}{dx} = \sec^2 x$ $v = \tan x$

$u = \sec^3 x$ $\frac{du}{dx} = 3\sec^2 x \times \sec x \tan x = 3\sec^3 x \tan x$

$$I = \sec^3 x \tan x - \int 3\sec^3 x \tan^2 x \, dx$$

$$= \sec^3 x \tan x - \int 3\sec^3 x (\sec^2 x - 1) \, dx$$

$$= \sec^3 x \tan x + \int 3\sec^3 x \, dx - \int 3\sec^5 x \, dx$$

$$4I = \sec^3 x \tan x + 3 \int \sec^3 x \, dx$$

$$I = \frac{1}{4} \sec^3 x \tan x + \frac{3}{8} \sec x \tan x + \frac{3}{8} \log(\sec x + \tan x) + C$$

d) i.e. $F_n^2 - F_{n+1}F_{n-1} = (-1)^{n+1}$

STEP A

Prove true for $n=2$

LHS $F_2^2 - F_3 \times F_1 = 1^2 - 2 \times 1$

RHS $(-1)^3 = (-1)$

$\therefore \text{LHS} = \text{RHS} \therefore \text{true for } n=2 \checkmark$

STEP B

Assume result true for $n=k$ ($k \geq 2$)

i.e. $F_k^2 - F_{k+1}F_{k-1} = (-1)^{k+1}$

Now prove true for $n=k+1$

i.e. $F_{k+1}^2 - F_{k+2}F_k = (-1)^{k+2}$

$$\begin{aligned} \text{LHS} &= F_{k+1}^2 - F_{k+2}F_k \\ &= F_{k+1}^2 - (F_{k+1} + F_k) \times F_k \\ &= F_{k+1}^2 - F_{k+1}F_k - F_k^2 \\ &= F_{k+1}(F_{k+1} - F_k) - F_k^2 \\ &= F_{k+1}F_{k-1} - F_k^2 \\ &= -(F_k^2 - F_{k+1}F_{k-1}) \\ &= -(-1)^{k+1} \\ &= (-1)^{k+2} \\ &= \text{RHS} \end{aligned}$$

using recursive definition
 $F_{k+2} = F_{k+1} + F_k$

Use of
Recursive
Definition

using recursive definition
rearranged
 $F_{k+1} - F_k = F_{k-1}$

using assumption / Use of
assumption

STEP C

By principle of mathematical induction, hypothesis
proven true for all integer $n \geq 2$.